

EUGENE CHO

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EDUCATION

University of California, Davis

Sep. 2026

B.S. in Computer Science

EXPERIENCE

Sealing Technologies, a Parsons Company

May 2026 – Present

Software Engineer I (formerly Software Engineer Intern)

Columbia, MD

- Re-architected networking and routing for a multi-service deployment after tracing deployment failures to hardcoded addresses and hostname rewrites; introduced hardened **NGINX** routing, browser-origin URL resolution, and standardized API ports
- Decomposed a single Compose pod into independently managed **systemd** services and automated provisioning with **Ansible**, enabling reproducible deployments and local/production parity across routing, TLS, and ports
- Designed and shipped context-window visibility for Zepharis AI, an air-gapped **React/FastAPI/vLLM** assistant, owning research, PRD, prototypes, and implementation; translated local-model limits into a context gauge, threshold banner, and inline tooltip

AggieWorks (UC Davis Product Organization)

Oct. 2024 – Jun. 2026

Technical Product Manager

Davis, CA

- Led a cross-functional team of engineers, designers, product managers, and marketers as a founding member of Davis Cattlelog, defining roadmap and product scope as the platform grew to **60K+** unique users and **9.4K+** monthly active users
- Served as primary engineer across the **React/TypeScript** frontend, **FastAPI/PostgreSQL** backend, ETL platform, and containerized infrastructure, authoring most of the codebase and sustaining development across successive student teams
- Built staging/production delivery and observability with **GitHub Actions**, **Docker Compose**, Cloudflare Tunnels, Prometheus, Grafana, and Loki, separating application, networking, and monitoring deployment workflows
- Re-architected semantic search and production performance using **pgvector** embeddings, fuzzy matching, response caching, and frontend virtualization, delivering **4×** faster retrieval, **90%** lower API latency, and **99.9%** uptime

Persist AI (YC W23)

Feb. 2026 – May 2026

Software Engineer Intern

Sacramento, CA

- Engineered a multi-agent web platform for pharmaceutical formulation workflows using **FastAPI**, **React**, and **Vite**, integrating **22** specialized backend agents into a unified system for project execution, collaboration, and research automation
- Built and maintained **37** backend API endpoints covering chat, session lifecycle, project CRUD, task management, job execution, snapshots, and visualization services, powering real-time agent-driven workflow orchestration and persistent project tracking
- Designed and supported real-time and asynchronous patterns across **27+** JavaScript and Python services, using WebSockets, SSE, background jobs, and multi-service orchestration to support long-running computations

Kaiser Permanente

Jun. 2025 – Dec. 2025

Software Engineer Intern

Remote

- Built and optimized event-driven backend services in **Spring**, **Express**, and **Kafka** to power an NLP-guided pre-appointment survey and smart triage system, streaming and processing over **500K** monthly unique sessions across patient intake workflows
- Led the end-to-end implementation of a School Note decisioning feature, integrating a **SOAP/XML** scaffold and a config-driven release agent pattern to improve scalability, observability, and maintainability
- Resolved a **Redis** cache miss pattern in patient revisit lookup by introducing time-bucketed cache keys and negative caching with TTLs, reducing external API calls by over **99%** and cutting median latency by **95%**

PNA Bio Inc.

Sep. 2024 – Jan. 2025

Software Engineer Intern

Newbury Park, CA

- Built and deployed customer-facing bioinformatics tools for molecular sequence analysis and PNA design using **PHP**, **Python**, and CGI scripts for a site serving **1.5K+** monthly visitors
- Automated sequence-design workflows through validated data-processing pipelines, reducing manual processing time by **90%**

PROJECTS

Evolution Atlas | React, TypeScript, D3, Node.js, Git

Aug. 2026 – Present

- Built a local-first React/D3 repository-history explorer that replays Git path operations across **2,693** commits into **43** reconstructable checkpoints, making codebase evolution traceable to commit-level evidence
- Architected a versioned Git-analysis pipeline that separates observed repository facts from AI-proposed interpretations, with deterministic validation for merges, renames, identity resolution, and evidence linking

TECHNICAL SKILLS

Languages: Python, Java, Go, JavaScript/TypeScript, SQL, C/C++, PHP

Frameworks & Libraries: React, Next.js, Node.js, Express, SvelteKit, FastAPI, Fiber, Spring Boot, vLLM

Infrastructure & Data: PostgreSQL, Redis, Kafka, Docker, Podman, Ansible, NGINX, systemd

Tools: Git, GitHub Actions, Jenkins, Splunk, Dynatrace, Linux

Awards: Eagle Scout Award, ACSL Intermediate Division Finalist, Ventura County Fire Department Citizen Award